



DESIGN MEETS NEUROSCIENCE: A PRELIMINARY REVIEW OF DESIGN RESEARCH USING NEUROSCIENCE TOOLS

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The past decade has witnessed a burgeoning interest in the intersection of neuroscience and design. Recent advancements in tools for measuring brain activity enable design researchers to investigate how brain function supports mental processes, complementing self-report and behavioral measures in exploring design cognition. With the aim of providing a preliminary understanding of how design researchers have approached neuroscience to date, we examined (1) previously explored neural research topics, (2) available tools and their contributions and limitations, and (3) the challenges of conducting design research using neuroscience tools. We start with a brief overview of major neuroscience tools measuring brain activation including electroencephalography (EEG), magnetic resonance imaging (MRI), near-infrared spectroscopy (NIRS), and positron emission tomography (PET). We then present a review of design research specifically on neural processes involved in design thinking and creativity. Consistencies of brain activity across studies are identified. Distinct patterns of brain activation associated with specific comparisons (e.g., design thinking vs. problem-solving) observed in previous studies are also summarized. An introductory review of design research using neuroscience tools on design evaluation and aesthetics is then provided. In addition to the promises of the fast-growing interdisciplinary collaboration of design and neuroscience, we outline technical, practical, and experimental difficulties caused by the limits of neuroscience tools for design research studies.

INTRODUCTION

BACKGROUND

Scientists have long been interested in the neuroscience behind the human mind. In the past, researchers were limited to indirect, clinical, or anatomical approaches for examining the structure and function of brains (see Sandrone et al., 2014). Now, there are a variety of technologies available such as electroencephalography (EEG), magnetic resonance imaging (MRI), near-infrared spectroscopy (NIRS), and positron emission tomography (PET). These tools are generally categorized into structural imaging and functional imaging methods (see Cacioppo et al., 2008). Structural imaging focuses on the anatomy of the brain, and functional imaging measures brain operations. Both support the understanding of brain activity associated with cognitive design processes.

Traditional methods (e.g., interview, observation) for investigating cognitive processes in design research are weak in capturing non-verbal thought (Cross, 2001). Self-report measures, which are based on what people say, are commonly used methods for investigating design cognition but may not reflect subconscious or unconscious reactions. The measurement of the individual brain and bodily functions provides a wider range of new tools for design research. Recent advancements in methodologies and instruments for measuring brain activity enable design researchers to more directly investigate how brain function supports such mental processes (Ball & Christensen, 2019). A previous review (Seitamaa-Hakkarainen et al., 2016) on the use of brain imaging methods in design studies indicated that cognitive neuroscience provides valuable tools for understanding the activities, skills, and cognition of design.

Gero and Milovanovic (2020) identified three paradigms for studying the cognitive processes of design: design cognition, design physiology, and design neurocognition, that can be measured through human minds, bodies, and brains. Individuals' minds may be investigated through methods such as interview; while galvanic skin response (GSR) and eye-tracking convey information about the function of the body (e.g., skin conductance, eye movement respectively); and neurocognitive tools (e.g., EEG, NIRS, and MRI) reveal the function and structure of the brain.

Physiological and brain measures, working to complement each other, provide a synthetic approach to capture human behavior. Physiological measures such as eye-tracking inform researchers about behaviors that can be associated with design activities performed by subjects. Eye-tracking technology enables the measurement of various eye metrics such as frequency and duration of fixation, the direction of the gaze, and pupil dilation. Studying eye activity provides insight into visual reasoning during design. For example, Lohmeyer et al. (2013) investigated visual strategies of designers with varying levels of expertise when analyzing design representations (e.g., sketches), and Cho and Su (2021) explored visual attentional pattern when looking at three-dimensional environment scenes with spatially relevant or decorative elements. Unlike physiological measures that focus on bodily response, neurocognitive measures examine brain activity. Increased use of neurocognitive tools contributes to an understanding of the design process on a neurological level (Zuanon, 2014). These tools provide new possibilities to experimentally trace the interrelations between brain activity and design cognition, thus complementing self-report (e.g., interview) and behavioral measures (e.g., observation) that are commonly used in design research. While acknowledging the contribution made by tools falling under design physiology and design cognition paradigms, we focus on neurocognitive tools for measuring brain activity because the application of neurocognitive methods in design research is a relatively new area in need of study.

APPLICATIONS TO INTERIOR DESIGN

The interdisciplinary approach promises to provide insights into human brain responses to design-related stimuli for various design fields such as interior design (see Bower et al., 2019). Neuroscience is an effective means of exploring the importance of various interior design elements in influencing human health and function, informing design researchers and practitioners about the neural activity associated with the complex interaction between humans and interior environments.

Interest in linking the study of interior design to the human experience inherent within neuroscience has surged recently (Karakas & Yildiz, 2020). A large portion of neuroscience research on interior design focuses on the effect of visual elements and characteristics of interior design, including form and geometry, color, light, ceiling height, materiality, furniture placement, style of interior furnishings (e.g., Banaei et al., 2017; Erkan, 2018; Ibrahim, 2019; Shin et al., 2015; Vartanian et al., 2015), and on health and well-being (Bower et al., 2019; Nanda et al., 2013). Neuroscience research addresses the influence of visual stimuli in the built environment on stress, restorativeness, emotions, and related concepts (Karakas & Yildiz, 2020). Apart from visual networks, interior design engages neural networks underpinning multiple sensory experiences including auditory, olfactory, and somatosensory, which evoke behavioral responses including approach and avoidance (Vartanian et al., 2015). Previous work (Coburn et al., 2017; Edelstein & Macagno, 2012) discussed ambient environmental factors such as sound as an environmental stressor, building temperature as an influencer on emotional state and comfort, and tactile interactions as contributors to spatial experience.

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OBJECTIVES

Although we focus on approaches for measuring brain activation, this is not a how-to paper instructing the use of neuroscientific methods nor are specific results provided for every

investigation cited. Rather, the objective is to review how design researchers have approached neuroscience to date by examining (1) previously explored neural research topics, (2) available tools and their contributions and limitations, and (3) the challenges of conducting design research using neuroscience tools. With the growing trend in using neuroscience approaches in design studies, another important purpose of this paper is to inform design researchers about what they should take into consideration before determining whether to use neuroscientific tools for their own research, the challenges for which they should be prepared, and the value/need for collaborative expertise to set up appropriate protocols and analyses.

A BRIEF OVERVIEW OF NEUROSCIENCE TOOLS FOR DESIGN RESEARCH

The use of tools originating from neuroscience can lead to a more comprehensive understanding of design activity (Lazar, 2018). In reviewing the relation of brain research methodologies to design research, Seitamaa-Hakkarainen et al. (2016) found that functional magnetic resonance imaging (fMRI) and EEG were the two primary neuroscience tools adopted in design research. In addition, PET and NIRS also may have an essential role in design studies. As shown in Table 1, these tools are included in this introductory discussion to provide a context for this body of neuroscience research and a background that helps designers to consider the value and limitations of each tool.

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One of the most important challenges researchers should acknowledge is the technical limitations of each neuroscience tool. Depending upon specific research questions, researchers need to make a methodological choice regarding spatial and temporal resolution, which are the determinants of neuroimaging performance. Spatial resolution refers to the capability a technique has to differentiate small structures, determining how

much detail we see accurately within the brain; temporal resolution refers to the ability to observe changes across time, determining how closely the measured activity matches the timing of the actual brain activity (Burle et al., 2015; Glover, 2011). Spatial resolution can tell exactly which brain region is active, while temporal resolution describes when the activation happened. Importantly, activation alone does not reveal if greater inhibition or excitation is involved. Hence, spatial and temporal mapping should be evaluated relative to other studies to interpret meaningfulness.

Lystad and Pollard (2009) presented a comparison table summarizing technical characteristics of neuroimaging techniques, including temporal and spatial resolution, and often there is a trade-off. fMRI is usually considered the hallmark of spatial resolution due to its capability to locate activation within the brain with millimeter precision. However, the temporal resolution of fMRI is poor so it cannot show the timing of the observed activation. EEG, on the other hand, has good temporal resolution, while its spatial resolution is poor.

ELECTROENCEPHALOGRAPHY (EEG)

EEG can be used to measure brain waves or event-related potentials—the averaged results of repeated brain responses to specific stimuli (Savoy, 2001). Non-invasive EEG methods place multiple electrodes on the scalp with conductive gel or more recently with dry electrodes. The development and basic principles of EEG technology as well as its applications have been well documented in a variety of sources (e.g., Binnie & Prior, 1994; Gevins, 2002). Due to its high temporal resolution, and with techniques applied to reveal neural spatial sources and functional associations, EEG continues to be an important monitoring and diagnostic tool in neurological disorders and to reveal normal brain function.

MAGNETIC RESONANCE IMAGING (MRI)

Lazar (2018) regarded neuroscience tools, MRI in particular, as an alternative tool for design studies that equips researchers with the ability to examine design cognition that cannot be verbalized in traditional methods. Seitamaa-Hakkarainen et al. (2016) suggested that MRI has not

Table 1. An overview of neuroscience tools for design research

	Description	Strengths	Limitations	Applications
Electroencephalography (EEG)	A non-invasive brain mapping technology for continuously recording electrical brain activity with millisecond temporal resolution	• Reveals both brain function and structure • Accurate temporal resolution and safety • Can be recorded while subjects are moving in real-world environments, virtual settings, and facing digital scenes • Portability of EEG equipment allows for the investigation to be performed in the natural working environments of designers • Widespread availability	• Cost (ranging from very low to very high equipment cost; but medium cost EEG is more affordable than functional magnetic resonance imaging) • Low-cost equipment often can only provide limited resolution • Spatial resolution of brain structures	• Functions support design cognition • Investigated the relationship between brain response and different design evaluations (e.g., Deng & Wang, 2019; Giraudet et al., 2015; Vieira et al., 2020) • Zhao et al. (2020) presented a task-related EEG framework for guiding studies on EEG-based cognitive and affective states of designers in the context of design
Magnetic resonance imaging (MRI)	An indirect measure of brain activity based on the increase in blood flow following neuronal activity, known as hemodynamic change (Huettel et al., 2004); the detected signal is dependent on the change in oxygenation (i.e., the blood oxygenation level dependent signal, BOLD)	• High spatial resolution and safety • Reveals both brain function and structure	• High equipment cost • Limited availability • Relatively poor temporal resolution, limited by the hemodynamic response time which occurs after neural signals evoke activity (Glover, 2011) • Machine noises: loud sound during scanning • Subjects must remain still, lying on a platform or in a tube	• Differentiated between creating and recalling original ideas and between creative generation and evaluation (Benedek et al., 2014; Ellamil et al., 2012) • Compared design with and without inspirational stimuli (Goucher-Lambert et al., 2019) • The effect of design training (e.g., designer vs. non-designer) (Kowatari et al., 2009) • Distinguished design thinking from problem-solving (Alexiou et al., 2009)
Positron emission tomography (PET)	A functional imaging technique based on the decay of radioactive atoms releasing positrons, utilizing the properties of the radiation generated when positrons are	• Sensitivity and flexibility • Relatively comfortable and quiet physical and psychophysical environment of scanning, thereby reducing	• Safety concerns about radiation exposure; some groups (e.g., children) cannot participate unless medically necessary • Injection with radioactive	• Activated brain areas involving solving creative tasks revealed brain correlates of creativity (Bechtereva et al., 2007; Bekhtereva

(Continues)

Table 1. Continued

	Description	Strengths	Limitations	Applications
	annihilated by interacting with electrons (Raichle, 1983)	claustrophobia, distraction from noise, and other psychophysical issues (Savoy, 2001)	substances limits frequency of use with any subject • Spatial resolution of brain structures (inferior to MRI) • Relatively poor temporal resolution compared to MRI and EEG • Longer duration of scans than MRI (Reiman et al., 2000) • Scarce availability • High operating cost (Lystad & Pollard, 2009)	et al., 2000; Starchenko et al., 2003) • Brain activation associated with visual imagery and high-level visual perception (Kosslyn et al., 1997), visual and mental spatial exploration (Mellet et al., 1995), and visual attention (Corbetta, 1998) • Emotional responses to varying olfactory and auditory stimuli (Royet et al., 2000)
Near-infrared spectroscopy (NIRS)	A non-invasive optical technology that enables functional imaging of brain activity and measures changes in the concentration of both oxy-hemoglobin (oxy-Hb) and deoxy-hemoglobin (deoxy-Hb) (Villringer & Chance, 1997)	• Flexibility and portability • No strict restrictions on motion • Low cost • Biochemical specificity (Schroeter et al., 2002) • More convenient and less expensive for studying brain function in practice and may be an appropriate substitute for MRI depending on the study topic and protocols	• Poor spatial resolution • Low signal-to-noise ratio (Cui et al., 2011) • Time delay compared to EEG methods • Practical difficulties including the quantification issue of NIRS data, the lack of standard methods for NIRS data analysis, and difficulties measuring deep brain structures (Hoshi, 2007)	• Shealy and colleagues conducted studies with a specific focus on design (see Hu et al., 2021; Shealy et al., 2020; Shealy & Gero, 2019) recording changes in students' brain activation during engineering design.

informed us directly about designer mental processes but has helped us analyze designer behaviors under specific conditions and trace brain activity associated with the cognitive processes of solving complex design problems. Detecting metabolic function over time, fMRI is based on MRI that image anatomical structure (Glover, 2011). MRI technology produces three dimensional pictures of brain structure, while fMRI generates pictures of metabolic activity occurring within the structures (Pilcher, 2004). fMRI uses MRI to assess hemodynamic change resulting from neuronal activity associated with certain brain area. Alexiou et al. (2009) and Gilbert et al. (2010) conducted fMRI studies where subjects were asked to perform tasks that portray a real-world interior design problem. These studies suggest that particular portions of the brain, in this case the right dorsolateral prefrontal cortex, are heavily involved in the generative phase of design cognition. Such brain localization findings provide valuable insights into design thinking from the perspective of neuroscience. The knowledge contributes to generating and validating hypotheses about the nature of design, pertaining to the neural network underlying specific behaviors

when addressing design tasks, which provides further evidence to differentiate design from non-design cognitive functions.

POSITRON EMISSION TOMOGRAPHY (PET)

Since the 1980s, researchers have used PET to image how brains work when subjects were completing tasks designed to manipulate specific mental processes and found that PET contributes to the localization of mental functions in the brain (Nutt, 2002). PET has played an important role in research on creative thinking and visual studies, as well as different sensory experiences. Bekhtereva et al. (2000) chose the use of PET over MRI for creativity research due to its higher possibility of reliable detection as detecting neural correlates of the delicate creative process is hard, and the inevitably noisy, uncomfortable MRI procedures makes the detection even harder.

NEAR-INFRARED SPECTROSCOPY (NIRS)

Originally developed for the clinical purpose of monitoring tissue oxygenation (Jöbsis, 1977), NIRS has also advanced as an effective tool for neuroimaging studies. A wide range of NIRS instruments have been designed for examining various brain activities, known as functional near-infrared spectroscopy (fNIRS). Using near-infrared light that can be transmitted over long distances through biological tissues to examine brain function, fNIRS shows increasing potential for use as a hemodynamic-based alternative to fMRI mostly due to its portability and greater tolerance for motion (Scarapicchia et al., 2017). Kopton and Kenning (2014) described the results of fNIRS studies for investigating economic decision-making and developed an assessment-criteria for determining whether to use fNIRS in neuroeconomic research. Several NIRS studies have explored the influences of different visual (i.e., pictures and imagery) and auditory (i.e., emotional sounds) stimuli on brain activation during participants' decision-making (Köchel et al., 2011; Plichta et al., 2011). These findings may be transferred to design studies on user preference and consumer response to product innovation. Further, Ferrari and Quaresima (2012) provided a brief history of fNIRS development and fields of application (e.g., neurology, psychiatry, psychology).

Since each tool has advantages and disadvantages, it is difficult to determine which one is preferred over the others. Regardless, some technologies are suited for certain types of research questions due to their technical characteristics, notably temporal and spatial resolution. According to an overview of functional neuroimaging techniques (Lystad & Pollard, 2009), with good spatial resolution, fMRI and PET are better suited to answering “where” questions, namely where in the brain (which specific brain area) is activated, while EEG are more appropriate to answering “when” questions, namely when exactly the brain activation happened, thanks to its excellent temporal resolution. Lystad and Pollard also concluded that fMRI appears to have largely replaced PET. In comparison, fNIRS has relatively poor spatial resolution and limited temporal resolution (comparable to that of fMRI) (Scarapicchia et al., 2017). However, recent advancements have developed portable fNIRS devices that allows for studying activities involving full-body movement such as walking (Holtzer et al., 2011) that would be impossible with fMRI.

In summary, no individual tool seems to be perfect; therefore, multimodal neuroimaging studies that combine multiple imaging modalities to generate comprehensive understandings of neural activity are increasingly used in neuroscience research (Liu et al., 2015). Further, neuroscience tools continue to improve as a result of technical advancements.

CHALLENGES FOR DESIGN RESEARCH USING NEUROSCIENCE TOOLS

Neuroscience can be extended to and grounded in design research. Integrating neuroscience tools with design research methodology is a fruitful area, providing important implications for design practice. However, in addition to technical limitations of each neuroscience tool pertaining to temporal and spatial resolution, there are feasibility issues and experimental difficulties.

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PRACTICAL DIFFICULTIES

Practical difficulties involved in applying neuroscience tools to design research include, but are not limited to, the accessibility of equipment and facility, operational expense, amount of training needed, and participant recruitment. While many universities house fMRI facilities, access to an fMRI scanner can be restricted due to the capacity of the scanner (e.g., the magnet field strength) and normally long wait times for fMRI scans affected by multiple factors including the number of scans and the number of hours of operation per machine per week. Moreover, the cost of purchasing an fMRI is very high, and the operational costs are expensive. Compared to fMRI, EEG is more feasible; it may be the most affordable neuroscience equipment. However, the training needed to proficiently use either tool, especially fMRI, is extensive.

Another challenge is participant recruitment. Neural studies often have more exclusion criteria than design studies using conventional self-report and behavioral measures. An fMRI scanner sends a strong magnetic field to gather images of the brain and any metals that react to magnets can interfere with the scan, thus the recruitment must eliminate people with pacemakers or any implanted magnetic device. Lengthy experimental procedures are another reason making participant recruitment even more difficult. Neuroscience experiments use rigorous protocols to achieve repeatability and reliability and are typically divided into multiple runs, plus preparation time, so the total duration can be long. Also, some participants will react negatively to the claustrophobic experience of being confined in the fMRI apparatus (Savoy, 2001).

EXPERIMENTAL CHALLENGES

As neuroscience experiments typically test how a given neurocognitive process responds when the subject performs a behavioral task, neuroimaging data alone is best supplemented by behavioral measures to reveal structural, functional, unconscious, and conscious responses to design elements. Aguirre (2014) noted that associating neural underpinnings with specific

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behavior requires careful study protocols. Given the multiple and co-existing factors interacting during any human behavior, there are challenges in creating reliable and valid experimental settings where a neurocognition process can be completely isolated through a behavioral manipulation.

As in any laboratory study, generalization can be problematic if the experimental tasks used in laboratory settings cannot be generalized to real-life design context. Similarly, generalizing the neuroimaging results from individuals to populations can be limited due to the significant individual differences in the complex pattern of folds comprising cortical surface (Aguirre, 2014). In

addition, there are various confounding variables due to individual differences such as those caused by medical states and drug effects.

Despite these challenges, neuroscience tools have been increasingly used to answer research questions involving the design process. The following sections present a review of growing applications of neuroscience tools in design research that yields a wide variety of informative results concerning design thinking, creativity, design evaluation, and aesthetics.

DESIGN RESEARCH USING NEUROSCIENCE TOOLS TO STUDY DESIGN THINKING AND CREATIVITY

Neuroscience tools have been increasingly applied to addressed design problems in various ways. This section presents a review of previous applications of neuroscience tools in design research on design thinking and creativity. Tools measuring design neurocognition provide insight into various aspects of design thinking such as design reasoning and design analysis (Gero & Milovanovic, 2020).

DESIGN THINKING STUDIES

Alexiou et al. (2009) conducted an fMRI study ($N = 18$) to understand the neurological basis of design thinking. They hypothesized that design activity could be distinguished from problem solving at the neural level and created an experiment where subjects were asked to perform two types of tasks (i.e., problem-solving vs. design) while in the fMRI scanner. The problem-solving task asked for an interior furniture arrangement based upon well-defined requirements (e.g., “the desk is under the window”), while the design task required subjects to arrange the same set of furniture so that “the room is functional and comfortable” (p. 629), which is considered open-ended and ill-defined. The results indicated the two tasks were accompanied with different patterns of brain activation and that design and problem solving involved distinct cognitive functions. Although the study is limited, their findings are interesting and can shed light on the fundamental assumption, supported by the design community, that design is a distinct discipline involving a distinct mode of thinking and “designerly ways of knowing” (Cross, 1982). To characterize design as a distinct cognitive phenomenon, we need evidence of the biological and/or neurological basis of design that differentiates design from other non-design high-level cognitive abilities—the problem-solving in this reviewed study. By comparing the identified brain activation for design versus problem-solving, the study concluded that understanding and addressing design tasks involved a more extensive neural network compared to resolving problem-solving tasks. Specifically, the involved network incorporates the dorsolateral prefrontal cortex, dorsal areas in the anterior cingulate cortex, and areas in the medial temporal lobe associated with language and semantic processing, and another region associated with voluntary eye saccades and gaze control. According to previous research on brain function (Gilbert & Burgess, 2008; Goel & Grafman, 1995), the prefrontal cortex is commonly involved in complex planning and problem-solving. The dorsolateral prefrontal cortex, in particular, is engaged in executive function (i.e., mental skills important for the cognitive control of behavior), working memory, as well as directed attention; the anterior cingulate plays an essential role in cognitive control and supports information processing in other brain areas (Miller & Cohen, 2001). Drawing on previous knowledge about the role of different brain areas, the authors attributed the more extensive brain network activated by design activities to the undertaking of semantic operations, evaluations of responses and representations, and to addressing conflicts as well as decision-making involved in design process. This is a key step toward understanding the distinct neurological basis underlying cognitive abilities associated with design.



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Kowatari et al. (2009) devised a design task conceptualizing a new pen in order to distinguish artistic creativity from verbal creativity. The authors point out that to design a new pen requires the use of mental imagery and spatial transformation, which can elicit brain activation in a frontoparietal network as reported in previous studies (e.g., Ishai et al., 2000). The score of originality for designs created by each subject were analyzed along with brain activity measured by fMRI. Consistent with previous findings that individuals with professional training in a specific domain show differences in brain activity compared to those without (Hanakawa et al., 2003; Ouchi et al., 2005), a comparison of students who were design majors ($N = 20$) versus non-design majors ($N = 20$) showed that in the group of design major subjects, the score of originality for design is correlated with the degree of dominance of the right prefrontal cortex over the left. In addition, this study predicted an effect of design training on the left parietal cortex, a region associated with general visuospatial processing, resulting in increased creativity. The findings provide valuable insight into the neurological mechanism under which professional trained designers produce more creative design outcomes than amateurs. Further, examining the effect of training on specific brain networks that lead to greater creativity helps design educators reflect on which design tasks should be issued to students for facilitating the dominance of certain brain area (i.e., right prefrontal cortex) closely linked to enhanced creative thinking.

Goucher-Lambert et al. (2019) explored the neurological processes underpinning creative design ideation ($N = 21$), with a specific focus on the influence of inspirational stimuli (e.g., analogies),

as a means to assist designers. The intent was to demonstrate that inspirational stimuli enhance the fluency of idea generation in terms of the quantity of ideas generated and response times. The use of fMRI revealed differences in brain activation networks supporting reasoning with and without inspirational stimuli. A key finding is that ideating with inspirational stimuli significantly involved several temporal brain regions that are known to be fundamental for semantic word processing, memory, and knowledge of objects and facts.

Since 2019, the *Design Science* journal has published several articles addressing design neuro-cognition research across neuroimaging modalities including fMRI, fNIRS, and EEG. The details of the published studies are summarized below in Table 2.

CREATIVITY STUDIES

Neuroscience research on creativity has flourished over the past 20 years (Sawyer, 2011). To delineate the availability of approaches to the neuroscientific study of creativity, a complex field critical in scientific, cultural, and pedagogical domains, Fink et al. (2007) examined emerging methodologies used for measuring creative thinking. Despite the difficulties in the operationalization of creative performance that have caused the comparative rareness of studies identifying brain mechanisms coupled with creative thinking, we suggest that neuroscience approaches that have expanded our knowledge about this mental ability domain should motivate research from other scientific fields. Thus, we present an overview of empirical studies by means of EEG, fMRI, NIRS, and PET, with a specific focus on the experimental tasks used for studying creative thinking in neuroscientific research settings. An important concluding remark made in Fink et al.'s (2007) article is that established experimental tasks that have been empirically tested in behavioral studies of creativity may not directly transfer into neuroscientific studies; comparatively less complicated experimental tasks have been adopted during the performance of some neurophysiological measurements. Regardless, we point out that the reviewed studies have provided insightful knowledge regarding brain correlates of creative thinking, extending our understanding about creativity acquired through traditional measures such as the Torrance Tests of Creative Thinking (Torrance, 1966). In addition, the review found that the creative ideation tasks predominately used verbal stimuli, suggesting future research should pay attention to the operationalization of non-verbal creativity through experimental tasks with non-verbal stimuli.

With respect to non-verbal creativity, Pidgeon et al. (2016) presented a systematic review of 26 neuroimaging studies (i.e., 7 fMRI and 19 EEG) on visual creativity involving the generation of mental visual imagery causing the production of creative visual forms. Consistent with established theories of creative cognition, the findings confirmed the contributions of prefrontal cortex and occipitotemporal regions to visual creative performance associated with enhanced visual processing, including visual imagery and visual image manipulation.

Using fMRI ($N = 42$) with visual stimuli, Benedek et al. (2018) explored the brain processes associated with the generation of original ideas about object uses and discriminated it from the recall of old ideas. They suggested that generating genuinely new ideas builds on similar brain processes as retrieving original ideas from memory, while new idea generation involves stronger activation of left the supramarginal gyrus, a part of the inferior parietal lobe crucial to the construction of novel representations, as demonstrated previously by Benedek et al. (2014).

Ellamil et al. (2012) used a new fMRI paradigm ($N = 15$) to separate and alternate components of the twofold creative process (i.e., creative generation and creative evaluation), a well-developed psychological definition (e.g., Campbell, 1960), at the neural level. The generative component of creativity facilitates the generation of novel ideas, and the evaluative component enables the evaluation of usefulness of the generated ideas. Distinct brain activation associated with the two processes were identified. Preferential recruitment of medial temporal lobe memory network is associated with the creative generation, while combined recruitment of executive network and default network and activation of the rostrolateral prefrontal cortex, insula, and temporopolar cortex is associated with creative evaluation; executive and default network

Table 2. A summary of neuroimaging studies published in the *Design Science* journal

Study	Method and participant	Objective/research question	Key findings
Fu et al. (2019)	fMRI ($N = 20$)	Whether and where design fixation (i.e., blind adherence to design example) can be detected in designers' brain activation during conceptual design?	fMRI results uncovered increased activation in brain area associated with visuospatial processing when comparing design ideation with and without example solution Activation was detected in the right inferior temporal gyrus, left middle occipital gyrus, and right superior parietal lobule regions Significant differences were found in activation between conditions (i.e., with and without example) The activation pattern demonstrates a shift in mental resources that can occur with design fixation
Hay et al. (2019)	fMRI ($N = 29$)	Explored neural basis underpinning product design engineering (PDE) ideation; compared brain activation when performing ideation tasks (open-ended vs. constrained) and contrast with imagery manipulation tasks	PDE ideation is associated with stronger activity in left cingulate gyrus A preliminary association with activation in the right superior temporal gyrus was found No significant differences between open-ended and constrained tasks were found
Vieira et al. (2020)	EEG ($N = 36$)	Investigated neurophysiological activations of designers from industrial design and mechanical engineering in ill-defined design tasks and well-defined problem-solving tasks; the tasks were adopted and extended from a previous fMRI study (Alexiou et al., 2009)	Significant differences were found in activations between the problem-solving and design tasks and between industrial designers and mechanical engineers, which supported previous findings that designing is distinct from problem-solving on the neurophysiological level and that mechanical engineers are less likely to change cognitive behavior in response to tasks with different levels of openness as compared to industrial designers
Shealy et al. (2020)	fNIRS ($N = 27$)	Examined how brainstorming, morphological analysis, and TRIZ (Russian acronym for the "Theory of Inventive Problem Solving" approach) changes the way specific brain regions are activated in the prefrontal cortex (PFC)	As compared to TRIZ, brainstorming and morphological analysis recruit more cognitive activation across the PFC The highest activation was found in the right dorsolateral PFC (DLPFC) and ventrolateral PFC, areas associated with divergent thinking and ill-defined problem-solving TRIZ leads to higher activation in the left DLPFC associated with convergent thinking and making judgments Morphological analysis and TRIZ can activate a greater number of brain regions that correlate when generating design concepts
Hu et al. (2021)	fNIRS ($N = 20$)	How does the year of educational training (freshmen vs. senior) in engineering influence students' ability to generate solutions? How do additional requirements of sustainability influence students' ability to generate solutions?	Without additional sustainability requirements, freshmen generated significantly more solutions than senior students Greater cortical activation in the dorsolateral prefrontal cortex associated with cognitive flexibility, and divergent and convergent thinking were found in freshmen Greater activation in the ventrolateral prefrontal cortex associated with uncertainty processing and self-reflection were found in senior students

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showed positive functional connectivity during task performance. The findings indicated that creative thinking involves unique neural processes normally not engaged in non-creative thinking that typically does not require the generation of novel ideas. The researchers asked participants to design book cover illustrations, using an fMRI-compatible drawing tablet, while switching between the two modes of thought in an fMRI scanner. A primary strength of this study is that, compared to previous studies in which participants were asked to think of their creative solutions during task performance (e.g., Kowatari et al., 2009), this experimental task mimicked a typical exercise in design programs through the use of a custom-built fMRI-compatible drawing tablet (Tam et al., 2011) enabling a more natural environment for design students to work on the design tasks while in an fMRI scanner.

With the goal of increasing reliability and validity of creativity assessment in neuroscience research, Benedek et al. (2019) reviewed a set of study design parameters (e.g., time on task, number of trials, the type of responses collected) summarized from creativity research using neuroscience methods and further examined how the task type and the use of different methods such as fMRI and EEG determine the study parameters used. Aiming to increase reliability of neuroscientific measurements and maintaining valid assessments, the review discussed advantages and disadvantages of each method and recommended the adaptations of experimental tasks suited for neuroscience assessment.

In this section, we discussed how researchers have approached neural correlates underlying design activities and creative thinking commonly involved in design process. Building upon the review of neuroscience studies on brain activities associated with design thinking and creativity in general, we next provide insight into how the use of neuroscience tools can be of help in design development as well as increasing aesthetic experience related to art and design.

DESIGN RESEARCH ON DESIGN EVALUATION AND AESTHETICS USING NEUROSCIENCE TOOLS

Neuroscience approaches used in design evaluation and aesthetic appreciation have shown great potential for providing implications for design practice. Accordingly, we present an introductory review of previous studies that have tested hypotheses related to human brain response to specific design elements and overall aesthetic experience.

DESIGN EVALUATION AND DEVELOPMENT

Following recent advancements in the emerging interdisciplinary field known as consumer neuroscience or neuromarketing that links the neuronal activity underlying consumer decision-making to marketing research (see Hubert & Kenning, 2008; Lin et al., 2018), the growing interest in understanding human evaluations of design has resulted in the application of neuroscience methods to design development (e.g., Schoen et al., 2018). Neuroscience approaches can be used to predict consumers' future choices (Telpaz et al., 2015) in that the behavioral and neural responses to different on-screen visual marketing stimuli (e.g., logo, TV ads, packaging) leads to the recognition of specific consumer preferences such as brand preference (Venkatraman et al., 2012). Similarly, by exploring brain response to design ideas, concepts, features, and projects using neuroimaging techniques, we can better comprehend preference for various characteristics of design, thereby obtaining a more complete picture of user response to design outputs. Neuroimaging methods, especially EEG and fMRI, have proven to be promising tools for developing user interface design, package design, product design, landscape architecture, architecture, interior design, and related fields.

User Interface Design

Giraudet et al. (2015) compared the behavioral and neural responses to different on-screen visual notification designs—the operational safety-critical alarm involving blinking colored text

displayed on the screen, featured in the human machine interface for Air Traffic Control. This study found that more salient visual design is associated with enhanced task performance as well as brain response to auditory alarms. Lee and Seo (2010) applied multiple bio-signal measurement techniques to the usability assessment of web design. The study measured individuals' emotional reactions to different web designs by means of EEG, indicating the use of bio-signals is an effective method for evaluating web design.

Package Design and Product Design

Stoll et al. (2008) used fMRI to explore the way the brain processes visual stimuli, such as package design, by measuring individuals' brain activity while they evaluate the attractiveness of consumer product packages. Consistent with the findings from consumer neuroscience, the findings showed that the level of attractiveness of the package can lead to different activation in brain areas associated with visual processing, which sheds light on the importance of aesthetic packaging in capturing attention from potential consumers.

Focusing on the impact of aesthetic emotion and judgment, an fMRI study identified the neural mechanisms underlying the appreciation of everyday designed products (Yeh et al., 2015). Deng and Wang (2019) investigated user responses to different shapes of products, proposing a new product design method for constructing visual elements based on user preference identified from the EEG experiment. Ma et al. (2018) adopted EEG to measure brain response to personalized T-shirt designs and discussed the relation of the response to user preference.

Landscape, Architecture, and Interior Design

Zhang et al. (2019) examined cortical differences in the appreciation of landscape design by scanning participants using fMRI while they were exposed to photographs of natural landscapes and landscape gardens. The fMRI study identified brain regions associated with emotional experiences involving a sense of reward resulting from landscape design.

John P. Eberhard was the founding president of The Academy of Neuroscience for Architecture, a nonprofit organization created in 2003 to connect neuroscience research to architectural design practice. Eberhard (2009a) described links between neuroscience and architecture, providing a brief case history of how neuroscience has impacted design and how design impacts the brain. He suggested that architectural design should respond properly to various aesthetic, functional, and cognitive needs of occupants and neuroscience can be a new knowledge base for architecture in understanding how design supports cognitive functions. For example, the impact of classroom design on the cognitive activities of students can be studied by means of neuroimaging to investigate underlying brain activities. Further, in his book *Brain Landscape*, Eberhard (2009b) discussed design hypotheses for future research offering linkages with potential neuroscience application and how architecture should address various needs of people with different brain conditions; for example, certain areas of the brain responsible for creating visual representation of the spatial surrounding environment are damaged in people with Alzheimer's disease. They need well-designed built environments that help reduce their symptomatic behaviors.

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A recent review (Karakas & Yildiz, 2020) of neuroscience research on the impact of the built environment on the human experience synthesized key theories and methodology, with a presentation of two tables summarizing theoretical and experimental studies focusing on research topics that have emerged at the intersection of neuroscience and architecture, together with major outcomes from the selected experimental studies. This review analyzed a group of neuroscience research techniques applicable to architectural research including neuroimaging tools (e.g., fMRI, EEG, PET), electrodermal activity measurements (e.g., GSR), and other important ways for studying the nervous system (e.g., heart rate, blood pressure, eye movements). Some

key concepts summarized from the reviewed articles included restorative and stress reduction effects of environments concerning human mental health, aesthetic judgment of features of the built environment (e.g., furniture arrangement, visual complexity, ambiances of interior spaces), navigation and wayfinding experience associated with characteristics of environmental design (e.g., urban spaces and forms, building shapes, and ceiling height), and visual attention and engagement associated with architectural spaces.

Bower et al. (2019) reviewed evidence on the impact of specific design characteristics (e.g., materiality or texture, style of interior furnishings, form, and geometry) of interior spaces on human emotion measured by neurophysiological techniques such as EEG, fMRI, fNIRS, and GSR. This systematic review on how design characteristics of the interior built environment affect neurophysiological response contributed to further research on how interior design can evoke emotional responses on a neurophysiological level. The resulting knowledge about the impact of design can be used to support mental health and wellbeing. The authors recommended the future application of neurophysiological measures to post-occupancy evaluations of the built environment.

Ibrahim (2019) outlined the relationship between interior design and neuroscience, with a specific focus on the interplay between space perception, memory, and emotion. The author proposed several design principles (i.e., enhanced planning and navigation, enriched interior spaces, and increased presence of nature features) concerning the integration of interior design and neuroscience and subsequently discussed the principles' influences on human performance. Also described were various design elements (e.g., form, color) falling under each principle.

Previous studies explored the impact of both visual (e.g., lighting, form) and non-visual (e.g., sound, orientation) elements and characteristics of interior design. Shin et al. (2015) examined environmental lighting conditions in a residential environment (i.e., a furnished living room), focusing on the influences of direct lighting versus indirect lighting on emotions and the accompanying brain activity using EEG, suggesting the important role played by ambient lighting in architectural spaces. A further conclusion was that EEG signals can be used as a biological marker of environmental alterations. Banaei et al. (2017) conducted an EEG study to investigate the impact of interior forms defined by specific parameters (e.g., scale, angle, geometry) on brain activity underlying the affective state. Clusters of form features were extracted from interior images of living rooms, which were then used to construct three-dimensional (3D) room models representing different interior forms experienced through virtual reality. A key finding of the study was that, compared to linear interior forms, the curvature geometries have a stronger impact on activity in the anterior cingulate cortex (ACC), a brain region involved in cognition and arousal and known to be associated with aesthetic judgment, which confirms previous results from an fMRI study (Vartanian et al., 2013) that aesthetic judgments of curvilinear contours activated the ACC over rectilinear contours.

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Components of interior design that involve both visual and non-visual (e.g., auditory) elements have been studied as well. Virtual reality interior scenes were used with EEG to test various wayfinding cues such as visual and auditory stimuli (Edelstein & Macagno, 2012). Erkan (2018) studied wayfinding behaviors in interior spaces using EEG and virtual reality technology and examined how design features (e.g., ceiling height, object placement) influence human wayfinding. The results suggested, apart from social and personal characteristics of people such as gender and age, wayfinding behaviors were affected by space plan, spatial orientation, and the height of space. The study provided implications for integrating effective wayfinding into the design of interior environments. Further, Rounds et al. (2020) examined the effects of urban design on landmark recognition, with a specific focus on the evaluation of salient architectural features. This EEG study investigated neural processes associated with landmark usage and recognition during landmark-based navigation tasks in a virtual urban setting, providing implications for developing better wayfinding systems in urban planning.

AESTHETIC APPRECIATION AND JUDGMENT

Neuroscience methods have been increasingly applied to the relatively young field within cognitive neuroscience known as neuroaesthetics, which refers to the study of aesthetic perception of visual art or any objects eliciting aesthetic judgments at the neurological level (see Cinzia & Vittorio, 2009). Neuroimaging tools have proven helpful in disclosing brain activity involved in aesthetic appreciation and, thus, are used to explain various aesthetic experiences. A meta-analysis of fMRI studies investigating the neural underpinnings of visual aesthetic experience suggests aesthetic-related neural responses to artwork are a feature of human cognition associated with a widely distributed neural network (Boccia et al., 2016). Moreover, there has been a wide range of neuroaesthetics research on paintings (Vartanian & Skov, 2014), music (Freitas et al., 2018), performing arts (Ticini et al., 2015), dance (Cross & Ticini, 2012), fashion (Simo-Serra et al., 2015), and movies (Boksem & Smidts, 2015).

In addition, neuroaesthetics research on color has been gaining attention. Kawasaki and Yamaguchi (2012) addressed the relationship between visual attention and subjective preferences for colors through the investigation of attention-related brain activation by means of EEG. The analysis of EEG data suggested the neural activities can be influenced by unconscious color preferences. Kim et al. (2012) presented an fMRI study on the brain activities related to color preference and further discussed how color preference can affect emotional response, showing different brain activation patterns in response to the most and least favorite colors identified through a questionnaire on subjective color preference. Such results inform designers about the potential impact of choosing colors in accordance with people's subjective color preferences on people's brain response and emotional status. Supporting preferences is critical to mental well-being.

CONCLUSION

To inform design researchers and practitioners about the application of neuroscience methods to design, we reviewed design studies using neuroscience tools specifically on design thinking and creativity where consistencies are found in brain activity. Activation in brain areas such as prefrontal cortex (especially the dorsolateral prefrontal cortex), anterior cingulate cortex, and parietal cortex, are commonly observed across studies. Moreover, this review summarized findings from previous studies about distinct patterns of brain activation associated with ill-defined design versus well-defined problem-solving, designers versus non-designers (or less experienced designers), visual versus verbal creativity, creating versus recalling original ideas, creative generation versus creative evaluation, and ideating with versus without additional stimuli (e.g., example solutions, inspirational analogies). Although questions remain, the accumulated findings offer an essential starting point that may inform the next stage of understanding, where individual differences can be identified in the context of design processes and experience. Future studies aiming at a more detailed account of how the brain supports design thinking and creativity might involve the combined application of multiple neuroscience tools in such a way as to overcome the technical limitations of individual tools (Fink et al., 2007; Liu et al., 2015). Additionally, given that most previous studies concerning design activities asked participants to imagine design solutions in their minds (e.g., Kowatari et al., 2009), devices compatible with neuroscience tools, such as the fMRI-compatible drawing tablet (Ellamil et al., 2012; Tam et al., 2011), allows for mimicking actual design activities involved in the design process (e.g., freehand drawing, computer-aided design, and prototype making). These should be further developed so that the research findings can be better generalized to real-life settings.

Acknowledging the overall growth in interdisciplinary design research using neuroscience tools, we also presented an introductory review of studies on design evaluation and aesthetics, where credible research outcomes have been produced. The applications of neuroscience methods to testing design solutions may have great potential for identifying implications for various design practices including: user interface design, package design, product design, landscape architecture, architecture, and interior design. As Eberhard (2009a, 2009b) suggested, design impacts brains and

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minds, and neuroscience enabling designers to examine human brains are increasingly influencing decision-making. The built environment, in particular, would benefit from neuroscience approaches (Eberhard, 2009a)—“if we expand the horizon for neuroscience, it would eventually result in a new knowledge base for architecture.” (p. 753). The new knowledge will then inform designers about how the design of hospital patient rooms can improve the recovery of patients, how the design of workplace (e.g., offices, laboratories) can facilitate collaboration, and why students in classrooms lit with natural light can earn higher test scores. In reviewing studies linking neuroscience methodology to knowledge of human responses to the built environment, we noticed a growing interest in the impact of both visual features such as building shape, ceiling height, lighting, and object placement, and non-visual features, such as sound, on well-being.

The present paper included a sample of current literature explaining what might happen when neuroscience tools are used to address questions of design research. The use of neuroscience tools enables design researchers to investigate the impact of design on human experience. For example, EEG can be used to examine how design features affect human wayfinding behavior (Erkan, 2018; Rounds et al., 2020) and how interior forms defined by parameters such as scale and angle influence affective state (Banaei et al., 2017). Further, specific features of interior design (e.g., furnishings style, texture) can evoke different emotional responses on a neural level, making neuroscience tools that reveal neurophysiological responses coupled with emotion useful in post-occupancy evaluations of the designed space (Bower et al., 2019).

A burgeoning interest in the intersection of neuroscience and design has led to a fast-growing demand for understanding the significance of neuroscience methodology, the available tools, and whether consistencies can be found across published studies. As an emerging tool for design research, neuroscience technology is powerful in capturing brain activity that cannot be measured by commonly used methods and complements self-report and behavioral measures. In this paper, we were interested in neuroimaging approaches that measured cognitive brain activation including PET, NIRS, EEG, and fMRI, with a specific focus on the latter two. Design researchers should be aware of the technical, practical, and experimental difficulties involved. Technical limitations, specifically the trade-off between the temporal and spatial resolution of each tool, as well as multiple practical difficulties (e.g., accessibility of equipment, operational expense, participant recruitment) are challenging. Creating reliable and valid experimental settings for using neuroscience approaches in design studies is complex, thus collaboration with neuroscientists is highly recommended along with effective ways to advance this domain of study. Knowledge of previous research and the challenges of neuroscience methodologies are critical to future design research.

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